Supplementary Information

Which screens for machine-written scholarly text survive an adaptive author? · Ook Lee

1. Full AUC matrix

AUC of every method under every condition, against 200 unattacked human papers (100 machine papers per generator and condition; the judge column for mixing has 99, 100 and 64 papers for Claude, GPT-4o and Llama). Conditions: none (orig); feature edit at effort 0.25 to 1.0; LLM rewrite; hybrid (rewrite then edit) at effort 0.5 and 1.0; sentence mixing at 30% and 50%; typo noise at 3% and 10%; second rewrite. '–' means not run.

Columns: Count = count screen (rates, dispersion); Count+clu. = with clustering features; Log-lik. = log-likelihood; Binoc. = Binoculars; Fast-DGPT = Fast-DetectGPT; Judge = Gemini 2.5 Pro.

Supplementary Table 1 | Claude papers.

Condition	Count	Count+clu.	Log-lik.	Log-rank	Entropy	Binoc.	Fast-DGPT	TF-IDF	RoBERTa	Judge
orig	0.78	0.80	0.56	0.59	–	0.67	0.94	0.93	0.96	0.82
edit0.25	0.75	0.74	0.52	0.55	–	0.67	0.92	0.93	0.95	–
edit0.5	0.71	0.69	0.49	0.52	–	0.66	0.91	0.93	0.94	–
edit0.75	0.70	0.66	0.47	0.50	–	0.65	0.89	0.93	0.93	–
edit1.0	0.67	0.62	0.46	0.49	–	0.65	0.90	0.92	0.92	0.83
rewrite	0.82	0.82	0.65	0.67	–	0.53	0.95	1.00	0.98	0.97
hybrid0.5	0.75	0.70	0.57	0.61	–	0.54	0.92	1.00	0.97	–
hybrid1.0	0.73	0.67	0.55	0.59	–	0.55	0.93	1.00	0.96	0.96
mix30	0.67	0.67	0.14	0.13	0.10	0.43	0.69	0.80	0.84	–
mix50	0.60	0.58	0.05	0.04	0.03	0.35	0.61	0.64	0.60	0.96
noise3	0.77	0.79	0.43	0.46	0.35	0.62	0.83	0.93	0.95	–
noise10	0.76	0.78	0.24	0.27	0.17	0.65	0.77	0.92	0.89	–
rewrite2	0.82	0.84	0.62	0.64	0.46	0.46	0.97	1.00	1.00	–

Supplementary Table 2 | GPT-4o papers.

Condition	Count	Count+clu.	Log-lik.	Log-rank	Entropy	Binoc.	Fast-DGPT	TF-IDF	RoBERTa	Judge
orig	0.85	0.89	0.93	0.95	–	0.83	1.00	0.98	0.99	0.95
edit0.25	0.82	0.86	0.91	0.94	–	0.83	1.00	0.98	0.99	–
edit0.5	0.79	0.82	0.90	0.93	–	0.83	1.00	0.98	0.99	–
edit0.75	0.76	0.79	0.89	0.92	–	0.83	1.00	0.98	0.98	–
edit1.0	0.71	0.76	0.88	0.92	–	0.83	1.00	0.98	0.98	0.98
rewrite	0.83	0.91	0.94	0.96	–	0.81	1.00	0.83	0.98	1.00
hybrid0.5	0.66	0.77	0.89	0.93	–	0.81	1.00	0.84	0.96	–
hybrid1.0	0.51	0.67	0.88	0.92	–	0.80	1.00	0.85	0.96	1.00
mix30	0.77	0.79	0.37	0.36	0.19	0.61	0.97	0.87	0.96	–
mix50	0.68	0.67	0.12	0.11	0.04	0.52	0.92	0.74	0.90	0.96
noise3	0.84	0.89	0.85	0.89	0.58	0.82	1.00	0.98	0.99	–
noise10	0.82	0.88	0.62	0.70	0.34	0.82	1.00	0.98	0.99	–
rewrite2	0.78	0.85	0.89	0.92	0.61	0.75	1.00	0.72	0.98	–

Supplementary Table 3 | Llama papers.

Condition	Count	Count+clu.	Log-lik.	Log-rank	Entropy	Binoc.	Fast-DGPT	TF-IDF	RoBERTa	Judge

orig	0.84	0.93	0.99	1.00	–	0.97	1.00	1.00	0.99	0.98
edit0.25	0.79	0.86	0.99	0.99	–	0.96	1.00	1.00	0.99	–
edit0.5	0.73	0.80	0.98	0.99	–	0.96	1.00	1.00	0.98	–
edit0.75	0.71	0.78	0.98	0.99	–	0.96	1.00	1.00	0.98	–
edit1.0	0.64	0.71	0.98	0.99	–	0.95	1.00	1.00	0.98	0.98
rewrite	0.82	0.92	0.99	1.00	–	0.96	1.00	1.00	1.00	1.00
hybrid0.5	0.72	0.80	0.98	0.99	–	0.96	1.00	1.00	0.99	–
hybrid1.0	0.65	0.72	0.98	0.99	–	0.95	1.00	1.00	0.98	1.00
mix30	0.75	0.81	0.57	0.57	0.28	0.73	1.00	0.94	0.91	–
mix50	0.66	0.66	0.23	0.22	0.07	0.61	0.98	0.82	0.83	1.00
noise3	0.84	0.93	0.98	0.99	0.84	0.96	1.00	1.00	0.99	–
noise10	0.83	0.92	0.92	0.96	0.63	0.95	1.00	1.00	0.98	–
rewrite2	0.81	0.91	0.97	0.98	0.76	0.92	1.00	1.00	1.00	–

2. Count screens: transfer between generators and human-only training

Supplementary Table 4 | AUC (95% bootstrap interval) by training and test generator. Folds are grouped by source paper.

Train → test	Rates	Dispersion	Both
gpt4o → gpt4o	0.864 [0.84, 0.89]	0.870 [0.84, 0.90]	0.945 [0.93, 0.96]
gpt4o → claude	0.745 [0.70, 0.79]	0.702 [0.65, 0.74]	0.771 [0.73, 0.81]
gpt4o → llama	0.680 [0.64, 0.72]	0.656 [0.61, 0.70]	0.711 [0.67, 0.75]
claude → gpt4o	0.769 [0.73, 0.80]	0.796 [0.76, 0.83]	0.861 [0.83, 0.89]
claude → claude	0.854 [0.81, 0.89]	0.684 [0.63, 0.73]	0.866 [0.83, 0.90]
claude → llama	0.775 [0.74, 0.81]	0.676 [0.64, 0.72]	0.824 [0.79, 0.86]
llama → gpt4o	0.752 [0.72, 0.78]	0.695 [0.65, 0.74]	0.830 [0.80, 0.86]
llama → claude	0.744 [0.70, 0.79]	0.591 [0.54, 0.64]	0.770 [0.73, 0.81]
llama → llama	0.909 [0.88, 0.93]	0.763 [0.72, 0.80]	0.935 [0.91, 0.96]

Supplementary Table 5 | Unseen generator. Trained on human papers and the other two generators.

Target	Baseline	Clustering	Difference
Claude	0.781 [0.74, 0.82]	0.810 [0.76, 0.85]	+0.029 [-0.003, +0.056]
GPT-4o	0.853 [0.82, 0.88]	0.886 [0.86, 0.91]	+0.032 [+0.015, +0.052]
Llama	0.792 [0.75, 0.83]	0.861 [0.84, 0.89]	+0.069 [+0.045, +0.093]

Supplementary Table 6 | Trained on human papers only (no machine examples).

Target	Rates	Dispersion	Both	Clustering
Claude	0.509 [0.46, 0.56]	0.526 [0.47, 0.58]	0.471 [0.42, 0.52]	0.421 [0.37, 0.48]
GPT-4o	0.447 [0.40, 0.49]	0.716 [0.68, 0.75]	0.620 [0.58, 0.66]	0.439 [0.39, 0.48]
Llama	0.684 [0.64, 0.72]	0.511 [0.46, 0.56]	0.663 [0.62, 0.70]	0.638 [0.59, 0.68]

3. Two-sided probability scores

Supplementary Table 7 | One- and two-sided scores under perturbation. Two-sided score = absolute deviation from the median of the 186 human negatives (the threshold uses the evaluation negatives, so values are optimistic).

Score	Generator	Condition	One-sided AUC	Two-sided AUC
Log-likelihood	Claude	mix30	0.14	0.73
Log-likelihood	Claude	mix50	0.05	0.90
Log-likelihood	Claude	noise10	0.24	0.64
Log-likelihood	GPT-4o	mix30	0.37	0.50
Log-likelihood	GPT-4o	mix50	0.12	0.75
Log-likelihood	GPT-4o	noise10	0.62	0.39
Entropy	Claude	mix30	0.10	0.82
Entropy	Claude	mix50	0.03	0.94
Entropy	Claude	noise10	0.17	0.73
Entropy	GPT-4o	mix30	0.19	0.65
Entropy	GPT-4o	mix50	0.04	0.91
Entropy	GPT-4o	noise10	0.34	0.51

4. Artifact control

Supplementary Table 8 | TF-IDF classifier variants (see Methods of the main text; script s10_artifact_control.py).

Protocol	Generator	Raw	Artifacts removed	Function words only
grouped-cv	Claude	1.00	1.00	1.00
grouped-cv	GPT-4o	1.00	1.00	1.00
grouped-cv	Llama	1.00	1.00	1.00
unseen-generator	Claude	0.85	0.74	0.66
unseen-generator	GPT-4o	1.00	1.00	1.00
unseen-generator	Llama	0.99	0.99	0.93

5. Procedure details

Generation: three prompt levels of increasing detail; each paper was written in four requests (introduction, methods, results, discussion) from the title and abstract of a sampled human paper (scripts s1 and s2). Judge prompt: see s5_judges.py. Attacks: see s7_attacks.py and s9_adaptive.py. Compute: the study used about US$78 of API calls and a rented 48 GB GPU for the published detectors and the RoBERTa classifier. All code and derived data are at https://doi.org/10.5281/zenodo.23664508.